

Chapter 15: Physics-Informed ML & Quantum Machine Learning (QML)

Chapter 14 closed by handing off to a different real cross-cutting question than any single molecular modality: how the physical laws that govern a system — conservation of mass, the Schrödinger equation — can be built directly into the learned models. Chapters 1-14 have each constructed from a different real angle. Both halves of that question share a common motivation this chapter states up front. Every model in this book so far has been either purely data-driven (a QSAR classifier, a graph neural network, a protein language model — Chapters 5-8) or purely physics-based (AutoDock Vina’s empirical scoring function, OpenMM’s classical force fields — Chapters 11-12); none has forced a neural network’s own output to satisfy a governing differential equation everywhere, and none has used a quantum-mechanical computing device as part of the model itself. **Physics-informed neural networks** (Section 15.1) close the first gap: they add a real governing equation’s residual directly into a neural network’s loss function, so the network is penalized for violating known physics even in the wide gaps between sparse experimental data. **Quantum machine learning** (Section 15.2) approaches the second gap from a different direction: real molecular electronic structure is itself a quantum-mechanical problem that scales exponentially on a classical computer, and hybrid quantum-classical algorithms are a real, active, NISQ-era (Preskill, 2018) attempt to compute it using an actual quantum processor for the exponentially-hard part and a classical computer for everything else. This chapter builds a real, working, small-scale example of each: a physics-informed network that solves a real pharmacokinetic system from sparse, noisy data (Section 15.1), and a real hybrid quantum-classical algorithm that computes real molecular ground-state energies (Section 15.2).

15.1 Physics-Informed Neural Networks (PINNs)

15.1.1 The problem: a purely data-driven model has no notion of physics between its data points

A plain feed-forward regressor trained to fit a handful of noisy observations has exactly one objective: minimize its error *at those observations*. Nothing in that objective constrains what the network does *between* them — in a densely sampled, low-noise regime this rarely matters, because nearby data points already pin down the function’s shape, but real experimental biophysical data is frequently neither dense nor low-noise. A real clinical pharmacokinetic (PK) study, for instance, might sample a patient’s plasma drug concentration at only 6-10 time points over 24 hours (each blood draw has a real clinical and financial cost), and each measurement carries real assay noise. A network free to interpolate those sparse points however it likes will often produce a plausible-looking but physically wrong curve — one that violates, for instance, the basic real requirement that a fixed, finite dose of drug cannot appear or vanish from the body faster than the body’s own absorption and elimination processes allow. **Physics-informed neural networks** (PINNs; Raissi, Perdikaris, & Karniadakis, 2019) directly fix this by adding a second, unsupervised term to the training loss: at any point in the input domain — not just where data exists — automatic differentiation computes the network’s own exact derivatives, and a real governing differential equation’s residual, evaluated from those derivatives, is penalized directly. The network is thereby forced to obey the known physics *everywhere*, using the data only to pin down which of the

physically-valid solutions is the real one. Karniadakis et al. (2021) give the general framework a fuller treatment; this section develops one concrete real instance in full.

15.1.2 A real biophysical conservation law: one-compartment oral pharmacokinetics

Chapter 5 §5.2 introduced ADMET profiling only at the level of a single scalar property prediction (will this compound clear too fast, will it cross a membrane); this section models the actual real *time-course* of a drug’s plasma concentration after an oral dose, governed by a real, textbook pharmacokinetic conservation law (Gibaldi & Perrier, 1982): a one-compartment model with first-order absorption. The administered dose starts entirely in a “depot” compartment (the gut lumen or an injection site) and is absorbed into the systemic (central) compartment at a real rate proportional to the amount remaining in the depot, while simultaneously being eliminated from the central compartment at a real rate proportional to its own current concentration — two coupled real mass-balance ordinary differential equations,

$$\frac{dA}{dt} = -k_a A, \quad \frac{dC}{dt} = \frac{k_a A}{V} - k_e C,$$

where $A(t)$ is the real amount of drug remaining in the depot compartment, $C(t)$ is the real drug concentration in the central (plasma) compartment, k_a and k_e are the real first-order absorption and elimination rate constants, and V is the real apparent volume of distribution. This system has a real, closed-form analytical solution — the **Bateman equation** (Bateman, 1910), originally derived for radioactive-decay chains and adopted directly into pharmacokinetics because the mathematics is identical:

$$C(t) = \frac{F \cdot \text{Dose} \cdot k_a}{V(k_a - k_e)} \left(e^{-k_e t} - e^{-k_a t} \right),$$

with F the real oral bioavailability fraction. This closed form is what makes the one-compartment model an unusually good real testbed for this section’s own PINN: it provides a real, independent, exact ground-truth curve — never shown to either model during training — for honest quantitative comparison, the same “compute a real, checkable reference before trusting a numerical method on it” discipline Chapter 13’s redocking-validation control and this book’s other hands-on projects have applied throughout.

15.1.3 The PINN loss function

The real PINN loss combines three real terms — a data term, a physics term, and an initial-condition term — computed from a single shared network $(\hat{A}(t; \theta), \hat{C}(t; \theta))$:

$$\mathcal{L}_{\text{data}} = \frac{1}{N_d} \sum_{i=1}^{N_d} (\hat{C}(t_i; \theta) - C_i^{\text{obs}})^2$$

$$\mathcal{L}_{\text{physics}} = \frac{1}{N_c} \sum_{j=1}^{N_c} \left[\left(\left. \frac{d\hat{A}}{dt} \right|_{t_j} + k_a \hat{A}(t_j) \right)^2 + \left(\left. \frac{d\hat{C}}{dt} \right|_{t_j} - \frac{k_a \hat{A}(t_j)}{V} + k_e \hat{C}(t_j) \right)^2 \right]$$

$$\mathcal{L}_{\text{ic}} = (\hat{A}(0) - \text{Dose})^2 + \hat{C}(0)^2, \quad \mathcal{L}(\theta) = \mathcal{L}_{\text{data}} + \lambda_{\text{phys}} \mathcal{L}_{\text{physics}} + \lambda_{\text{ic}} \mathcal{L}_{\text{ic}}.$$

$\mathcal{L}_{\text{data}}$ is evaluated only at the N_d real, sparse observation times; $\mathcal{L}_{\text{physics}}$ is evaluated at N_c **collocation points** — unlabeled times sampled densely across the full time horizon, at which the ODE residual is computed directly from $d\hat{A}/dt$ and $d\hat{C}/dt$, obtained by real, exact automatic differentiation of the network’s own output with respect to its input (`torch.autograd.grad`), not a finite-difference approximation; and $\mathcal{L}_{\text{ic}}$ enforces the real initial condition that the entire administered dose starts in the depot compartment and none has yet reached the plasma — the literal, concrete instance of “incorporating biophysical conservation laws into loss functions” this section’s outline names, since $\hat{A}(0) = \text{Dose}$ is exactly a statement of real mass conservation at $t = 0$.

```
a_col, c_col = model(tau_col) # unlabeled collocation points
da_dtau = torch.autograd.grad(a_col, tau_col,
grad_outputs=torch.ones_like(a_col), create_graph=True)[0]
dc_dtau = torch.autograd.grad(c_col, tau_col,
grad_outputs=torch.ones_like(c_col), create_graph=True)[0]
res_a = da_dtau - (-KA_TRUE * T_SCALE * a_col)
res_c = dc_dtau - (T_SCALE * (KA_TRUE * a_col - KE_TRUE * c_col))
physics_loss = torch.mean(res_a**2) + torch.mean(res_c**2)
```

15.1.4 Real, hands-on demonstration: PINN vs. a plain regressor on sparse, noisy PK data

The project code lives in this chapter’s folder (`ch15_pinn_qml/pinn_qml.py`). A real one-compartment PK system ($k_a = 1.2 \text{ h}^{-1}$, $k_e = 0.3 \text{ h}^{-1}$, $V = 30 \text{ L}$, a 500 mg oral dose — plausible values for a fast-absorbed, moderately-cleared small molecule, not any specific marketed drug) is sampled at eight real, realistically-spaced clinical time points ($t = 0.25, 0.5, 1, 2, 4, 8, 12, 24 \text{ h}$), with 5% real Gaussian assay noise added to each. A plain feed-forward regressor and the physics-informed network above (identical architecture — a $1 \rightarrow 32 \rightarrow 32 \rightarrow 2$ MLP — so any performance difference reflects the physics term, not model capacity) are each trained on these same eight points and compared against the real, noiseless Bateman-equation curve over the full 24-hour window:

Model	RMSE, full curve (mg/L)	RMSE, $t < 1 \text{ h}$ (mg/L)	RMSE, $t > 12 \text{ h}$ (mg/L)
Plain regressor (no physics)	0.556	0.837	0.078
PINN	0.128	0.262	0.049

The PINN’s real error is **4.4× lower** than the plain regressor’s over the full curve, and **3.2× lower** specifically in the sparse, high-curvature early-absorption window ($t < 1 \text{ h}$) — exactly the region where two of the eight observations ($t = 0.25 \text{ h}$ and $t = 0.5 \text{ h}$) must, on their own, pin down the entire rising edge and the concentration peak, a genuinely hard interpolation problem for a method with no notion of the underlying pharmacology. The plain regressor, free to fit *any* smooth curve through those two points, systematically undershoots the true peak; the PINN’s

collocation-point physics loss forces its interpolation to already look like a real one-compartment absorption/elimination curve between the sparse observations, so it needs the data only to fix the *scale* and *timing* of that real shape rather than to discover the shape itself. The gap narrows, but does not close, in the elimination-phase tail ($t > 12$ h, mostly log-linear decay, an easier real interpolation problem the plain regressor already handles reasonably). At convergence (12,000 Adam iterations, ~ 101 s of CPU wall-clock time), the PINN’s own real loss terms confirm both objectives were actually satisfied, not merely traded off against each other: final data loss 1.9×10^{-4} (dimensionless, normalized units), final physics-residual loss 9.2×10^{-3} , final initial-condition loss 8.8×10^{-4} .

One real, disclosed data artifact, left uncorrected rather than quietly cleaned up: the $t = 24$ h synthetic observation is -0.063 mg/L — a real, physically impossible negative concentration, produced because the true concentration at that late timepoint (≈ 0.06 mg/L) is already close to the assay’s own noise floor, and unconstrained Gaussian noise can push a near-zero true value slightly negative. This is a genuine, common real-world limit-of-quantification effect (a real assay would instead report “below the limit of quantification” rather than a negative number), included here deliberately rather than filtered out, since a real physics-informed model should be robust to it — and both models above were trained on the literal, uncorrected value.

15.1.5 A harder real extension: inverse parameter estimation, and an honest identifiability caveat

Section 15.1.4’s PINN treats k_a , k_e , and V as known quantities and solves the *forward* problem (find the concentration curve consistent with both the physics and the sparse data). A real, harder, and arguably more practically valuable extension treats k_a and k_e themselves as trainable parameters, optimized jointly with the network — an *inverse* problem: recovering a compound’s own real pharmacokinetic rate constants directly from sparse concentration data via the physics-informed loss, rather than by traditional nonlinear least-squares curve fitting. This is a real, legitimate use case (early PK parameter estimation from limited sampling), but it runs directly into a genuine, well-documented pharmacological identifiability issue sometimes called **flip-flop kinetics**: when absorption is not substantially faster than elimination, the observed concentration-time *shape* alone can be close to equally well explained by two different (k_a, k_e) pairs with the roles of the two rate constants effectively exchanged, so the inverse problem can have (near-)degenerate solutions that no amount of additional physics-loss weighting resolves on its own — the physics constrains the model to a *valid* one-compartment curve, but does not by itself disambiguate which of two physically valid parameter sets generated it. A preliminary run of this section’s own inverse variant (not included in `pin_n_qml.py`’s committed results, to keep this chapter’s own reported numbers to what was fully, honestly converged) reproduced exactly this failure mode rather than recovering $k_a = 1.2$, $k_e = 0.3 \text{ h}^{-1}$ cleanly. Resolving it in general requires either a substantially better-conditioned loss (a real, nontrivial engineering problem, out of this chapter’s scope) or additional real information — most directly, knowing from independent evidence (e.g. an intravenous dosing arm) which rate constant is the faster one — disclosed here as a genuine open direction rather than a solved problem.

15.2 Quantum Machine Learning (QML)

15.2.1 Why electronic structure is a genuinely quantum problem

Every classical electronic-structure method this book has used or discussed so far — the force fields of Chapter 12, the neural network potentials (ANI-2x) that replaced them — is, at its core, an *approximation* built to avoid solving the real many-electron Schrödinger equation directly, because doing so exactly on a classical computer requires representing a wavefunction whose size scales **exponentially** with the number of electrons: a full configuration-interaction (exact) treatment of N spin orbitals requires storing amplitudes over a Hilbert space of dimension that grows as $\binom{2N}{n_{\text{electrons}}}$, intractable on any classical computer beyond a handful of atoms in a minimal basis. This is not a software-engineering limitation any faster classical hardware resolves — it is the reason quantum chemistry has spent decades building *approximate* classical methods (Hartree-Fock, density functional theory, coupled cluster) instead. A real, fault-tolerant quantum computer would not have this problem: a system of N qubits natively represents a 2^N -dimensional quantum state, so a molecular electronic wavefunction that would require exponential classical memory maps onto a *linear* number of qubits — a real, theoretically well-established complexity argument (Cao et al., 2019; McArdle et al., 2020) that motivates quantum computation as a fundamentally different approach to electronic structure, not merely a faster one. Today’s quantum hardware is not fault-tolerant, however — it is small, noisy, and error-prone (the “NISQ” Noisy Intermediate-Scale Quantum, era; Preskill, 2018) — which is exactly why **hybrid quantum-classical algorithms**, which keep the noise-sensitive quantum computation short and offload the optimization to a classical computer, are the real, practical near-term approach this section demonstrates.

15.2.2 The Variational Quantum Eigensolver (VQE)

The **Variational Quantum Eigensolver** (VQE; Peruzzo et al., 2014) is the canonical hybrid quantum-classical algorithm for exactly this problem. A molecule’s real second-quantized electronic Hamiltonian is first mapped onto a sum of Pauli-operator strings acting on qubits (the Jordan-Wigner transformation, used throughout this section); a parameterized quantum circuit — an **ansatz** — then prepares a real trial quantum state $|\psi(\theta)\rangle$ on the (real or simulated) quantum device, from a Hartree-Fock reference state augmented with single- and double-electron excitations (the same UCCSD family of excitations that underlies classical coupled-cluster theory). The real **variational principle** guarantees that the expectation value of the true Hamiltonian in *any* trial state upper-bounds the true ground state energy E_0 ,

$$E(\theta) = \langle \psi(\theta) | \hat{H} | \psi(\theta) \rangle \geq E_0 \quad \text{for all } \theta,$$

so a classical optimizer minimizing $E(\theta)$ over the circuit’s own real parameters — using only the *measured* expectation value from the quantum device at each step, not any assumption about the Hamiltonian’s structure — is guaranteed, at convergence, to have found a real, valid upper bound on the true ground-state energy, tight exactly when the ansatz is expressive enough to represent it. This inequality is not merely theoretical: Section 15.2.4’s own test suite verifies it directly against this chapter’s own real, converged VQE run.

15.2.3 A real, disclosed tooling substitution: PennyLane’s own differentiable Hartree-Fock backend

Building a molecular qubit Hamiltonian ordinarily requires an external quantum-chemistry package (PySCF or Psi4) to compute the one- and two-electron integrals that define it. This chapter’s own Windows authoring environment has no prebuilt PySCF wheel — installing it requires compiling from C/C++ source via `cmake/nmake`, a toolchain this environment does not have, the same category of real, disclosed installation barrier Chapter 11 hit with DiffDock’s conda-only distribution and Chapter 13 hit with AutoDock Vina’s Windows wheel gap. PennyLane (Bergholm et al., 2018) — chosen as this chapter’s own quantum computing library specifically because of this constraint — ships its own **differentiable Hartree-Fock** backend (`qchem.molecular_hamiltonian(..., method="dhf")`), computing the same real one- and two-electron integrals natively in Python with no external quantum-chemistry dependency at all. This chapter’s entire real VQE pipeline runs on `pennylane` and `torch/numpy` alone.

15.2.4 Real, hands-on demonstration: VQE ground-state energies for H2 and LiH

The project code lives in `ch15_pinn_qml/pinn_qml.py`. Two real molecules, at real, plausible equilibrium-region geometries, are run through the full real pipeline — building the real qubit Hamiltonian, running the real VQE, and independently checking the result against real exact diagonalization of that identical Hamiltonian (this chapter’s own real ground truth, not a literature number):

Molecule	Qubits	HF energy (Hartree)	VQE energy (Hartree)	Exact energy (Hartree)	Error vs. exact	Iterations	Wall time
H2	4	-1.117349	-1.136189	-1.136189	2.5×10^{-9}	18	1.6 s
LiH (active space, 2e/3o)	6	-7.863267	-7.864316	-7.864316	1.5×10^{-7}	92	45.0 s

Both real VQE runs converge to within machine-precision-level agreement with exact diagonalization — H2’s final error is 2.5×10^{-9} Hartree, LiH’s is 1.5×10^{-7} Hartree, both many orders of magnitude below “chemical accuracy” ($1 \text{ kcal/mol} \approx 1.6 \times 10^{-3}$ Hartree), because for a system this small the singles-and-doubles ansatz is essentially exact (it spans the same excitation manifold full configuration interaction does, for a 2-electron-in-3-orbital active space) — a real, honest reflection of how small these demonstration systems are, not evidence that VQE generally reaches machine precision on larger, real drug-relevant molecules. H2’s real correlation energy — the gap between the mean-field Hartree-Fock energy and the true, correlated ground state — is 0.0188 Hartree ($\approx 11.8 \text{ mHartree} \approx 7.4 \text{ kcal/mol}$), recovered in full by the 18-iteration VQE optimization from only 3 real variational parameters (one single and one double excitation amplitude, by symmetry). LiH’s active-space reduction (2 active electrons in 3 active orbitals, chosen to keep the qubit count and optimization wall time Colab-CPU-tractable — a real, disclosed compute-budget

choice, not a hidden approximation) uses 6 qubits and 8 variational parameters, converging in 92 iterations.

An honest scaling disclosure. Both real demonstrations above sit enormously far below the scale of any real drug-discovery-relevant electronic-structure problem. A druglike small molecule in even a modest basis set requires hundreds of spin orbitals, not four to six — directly translating, under today’s qubit-per-spin-orbital mapping strategies, to hundreds of qubits (well beyond both current NISQ hardware’s usable, low-error qubit counts and classical simulators’ own exponential memory wall, which is exactly why this section could only classically *simulate* a handful of qubits at all). This is the real, current state of quantum computational chemistry, honestly reported rather than glossed over: the theoretical exponential advantage of Section 15.2.1 is real, but realizing it for drug-discovery-scale electronic structure remains a genuine open research problem (Cao et al., 2019; McArdle et al., 2020), not a capability this chapter’s own two-molecule demonstration should be read as already delivering.

Limitations and what comes next

Both hands-on demonstrations in this chapter are deliberately small, real, and fully validated against an independent ground truth computed within the same script — not fabricated, not borrowed from a secondary source, and honestly disclosed where they fall short: the PINN’s own inverse-parameter-estimation extension runs directly into a genuine flip-flop-kinetics identifiability limit (Section 15.1.5), and the VQE demonstration’s two molecules are many orders of magnitude smaller than any real drug-discovery electronic-structure target (Section 15.2.4). Both limitations are the real state of each field today, not an artifact of this chapter’s own implementation choices. Chapter 16 shifts the book’s focus once more, from individual methods explored chapter by chapter to the first of two capstone projects that chain many of them together into a single, real, end-to-end small-molecule discovery pipeline.

References

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All PINN and VQE numbers cited in Sections 15.1.4 and 15.2.4 were computed directly by running `pinn_qml.py` on 2026-08-22, not taken from a secondary source — see `results/pinn_qml_results.json` to reproduce.